

Salesforce Summer Program – Day 11 (4–5 Hours)

☐ Goal for Today

Until now you learned how to:

- build systems
- automate workflows
- create UI
- integrate components

Today you will learn:

“How enterprise systems become reliable and scalable.”

Focus:

- Testing
- Async processing
- Reliability
- Background jobs

☐ Deep Learning Modules (IMPORTANT)

1 ☐ Apex Testing

Focus:

- Why testing matters
- Unit tests
- Preventing bugs
- Reliability

☐ VERY important in enterprise systems.

2 ☐ Asynchronous Apex

Focus:

- Future methods
- Queueable Apex
- Background processing
- Long-running jobs

☐ Understand conceptually. No need for deep syntax mastery.

☐ Light Completion Modules

3 ☐ Secure Server-Side Development (if pending)

4 ☐ Search Solution Basics (if pending)

☐ Videos (Verified Working Links)

1 ☐ Apex Testing Explained

https://www.youtube.com/watch?v=nMyKE6_U5qo

Focus:

- Test classes
 - Enterprise reliability
 - Why Salesforce requires tests
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2 ☐ What is Apex Testing?

<https://www.youtube.com/watch?v=RK7uSw1ynLI>

Focus:

- Unit testing basics
 - Business reliability
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3 ☐ Asynchronous Apex Explained

<https://www.youtube.com/watch?v=Lex6jrtB0Ho>

Focus:

- Background processing
 - Queueable jobs
 - Async logic
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☐ **CORE TASKS (MANDATORY)**

Task 1 ☐ Testing Thinking

Take your College Management System.

Suggest:

- 10 important test cases

Examples:

- Invalid email
- Duplicate registration
- Seats exceeding limit
- Attendance below threshold
- Notification failure

☐ Explain what problem each test prevents.

Task 2 ☐ Async Thinking

Give 5 examples where:

☐ background processing is better than immediate processing.

Examples:

- Bulk emails
 - Report generation
 - Large data import
 - Notifications
 - External synchronization
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Task 3 Reliability Thinking

Suppose your system crashes during:

- student registration
- payment update
- attendance update

What problems might happen?

How can testing help?

Task 4 Reflection Task

Answer:

- ☐ Why do enterprise systems require:
 - testing
 - scalability
 - async processinginstead of simple direct execution?
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☐ GitHub Submission

Create:

/day11-testing-async/

README.md must include:

- Why testing matters
 - What is asynchronous processing
 - Important test cases
 - Async use cases
 - Reliability discussion
 - Reflection
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☐ Revision Questions

1. Why is testing important?
 2. What problems happen without testing?
 3. Difference between synchronous and asynchronous execution?
 4. Why do enterprise systems use background jobs?
 5. Why should developers think about scalability?
 6. Why are test cases important?
 7. What happens when systems fail partially?
 8. Why do large systems require reliability engineering?
 9. Why should enterprise software avoid blocking operations?
 10. Why is enterprise software different from small scripts?
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☐ **End of Day Outcome**

You should now clearly understand:

- ☐ Why enterprise systems require testing
 - ☐ Why async processing exists
 - ☐ Reliability and scalability thinking
 - ☐ Importance of background processing
 - ☐ Enterprise-quality software mindset
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